汉字的故事

从甲骨文到拼音输入法

How Chinese Characters Work

From oracle bones to pinyin input

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Footnotes mark names, festival terms and words above HSK 4 (HSK 3.0 lists: github.com/ivankra/hsk30, MIT).

1 很多外国人第一次看到汉字的时候，都觉得它们像一幅[1]幅小画。汉字和字母不一样：一个字母一般只代表声音，一个汉字却常常同时带着声音和意思。汉字已经有三千多年的历史，是世界上一直用到今天的最古老[2]的文字之一。下面，我们就来看一看汉字是怎么来的，它们有什么规律，现代人又是怎么学习和使用汉字的。

最早的汉字

2 现在能看到的最早的成系统的汉字，是三千多年前商朝[3]的甲骨文[4]。甲骨文是刻在龟甲[5]和牛骨头上的文字。当时的国王[6]做很多事情以前，都要问一问祖先[7]和神[8]的意见：明天会不会下雨？这次打仗[9]能不能赢？今年的粮食会不会好？

1 幅 *fú* (measure word for pictures)
2 古老 *gǔlǎo* ancient
3 商朝 *Shāngcháo* the Shang dynasty
4 甲骨文 *jiǎgǔwén* oracle-bone script
5 龟甲 *guījiǎ* tortoise shell
6 国王 *guówáng* king
7 祖先 *zǔxiān* ancestors
8 神 *shén* gods
9 打仗 *dǎzhàng* to go to war

Many foreigners who see Chinese characters for the first time feel that they look like little pictures. Characters are different from letters: a letter usually stands only for a sound, while a character often carries both a sound and a meaning. They have a history of more than three thousand years, and they are one of the oldest writing systems in the world still in use today. Below we look at where characters came from, what patterns they follow, and how people today learn and use them. 1

The Earliest Characters

The earliest systematic Chinese writing we can see today is the oracle-bone script of the Shang dynasty, more than three thousand years ago. It was carved on tortoise shells (龟甲) and ox bones; hence the name 甲骨文, "shell-and-bone writing". Before doing many things, the kings of that time would ask their ancestors and the gods for advice: Will it rain tomorrow? Will this war be won? Will this year's harvest be good? 2

人们在骨头上用火烧，根据出现的裂纹[1]来判断好坏，再把问题和结果刻在骨头上。

这些甲骨[2]在地下埋[3]了三千多年。据说，1899年，一位学者[4]在买来的中药[5]里发现，一些叫做“龙骨[6]”的骨头上有奇怪的符号。他认出这是一种很古老[7]的文字，甲骨文[8]这才被人们重新发现。后来，考古学家在河南[9]安阳[10]挖[11]出了十几万片甲骨。上面一共有四千多个不同的字，到今天，大约有三分之一已经被认[12]出来了。

甲骨文以后，汉字的样子又变了很多次。商朝[13]和周朝[14]的人把字铸[15]在青铜器[16]上，叫“金文[17]”。

1 裂纹 *lièwén* crack
2 甲骨 *jiǎgǔ* oracle bones
3 埋 *mái* to bury
4 学者 *xuézhě* scholar
5 中药 *zhōngyào* Chinese medicine
6 龙骨 *lónggǔ* “dragon bones” (sold as medicine)
7 古老 *gǔlǎo* ancient
8 甲骨文 *jiǎgǔwén* oracle-bone script
9 河南 *Hénán* Henan (province)
10 安阳 *Ānyáng* Anyang, site of the last Shang capital
11 挖 *wā* to dig
12 认 *rèn* to recognize
13 商朝 *Shāngcháo* the Shang dynasty
14 周朝 *Zhōucháo* the Zhou dynasty
15 铸 *zhù* to cast (metal)
16 青铜器 *qīngtóngqì* bronze vessels
17 金文 *jīnwén* bronze inscriptions

People applied fire to the bones, judged good or bad fortune from the cracks that appeared, and then carved the question and the result onto the bone. 2

These oracle bones lay buried underground for more than three thousand years. It is said that in 1899 a scholar noticed strange symbols on some so-called "dragon bones" in the Chinese medicine he had bought. He recognized them as a very ancient form of writing, and that is how the oracle-bone script was rediscovered. Later, archaeologists dug up more than a hundred thousand pieces at Anyang in Henan. Altogether they contain over four thousand different characters, and about a third of them have been deciphered so far. 3

After the oracle-bone script, the shape of the characters changed many more times. People of the Shang and Zhou dynasties cast characters onto bronze vessels; these are called "bronze inscriptions". 4

4 秦始皇[1]统一中国以后，让全国都用一种文字，就是“小篆[2]”。在秦朝[3]和汉朝[4]的时候，人们为了写得更快，慢慢把弯弯的线条[5]变成了平直的笔画[6]，这就是“隶书[7]”。后来又出现了楷书[8]，它的样子一直用到了今天，我们在书上和手机上看到的字，基本上都是楷书。

看着东西画字

5 最早的一批汉字，很多是“象形字[9]”，就是照着东西的样子画出来的字。比如“日”，最早是一个圆圈，中间有一个点；“月”是一个弯弯的月亮；“山”像三座连在一起的山峰[10]；“人”像一个侧[11]着身子[12]站着的人。

1 秦始皇 *Qín Shǐhuáng* Qin Shi Huang, the First Emperor
2 小篆 *xiǎozhuàn* small seal script
3 秦朝 *Qíncháo* the Qin dynasty
4 汉朝 *Hàncháo* the Han dynasty
5 线条 *xiàntiáo* lines
6 笔画 *bǐhuà* stroke (of a character)
7 隶书 *lìshū* clerical script
8 楷书 *kǎishū* regular script
9 象形字 *xiàngxíngzì* pictograph
10 山峰 *shānfēng* mountain peak
11 侧 *cè* sideways
12 身子 *shēnzi* body

After Qin Shi Huang unified China, he had the whole country use a single script, the "small seal script". In the Qin and Han dynasties, to write faster, people gradually turned the curved lines into flat, straight strokes; this is "clerical script". Later came regular script, whose form is still used today: the characters we see in books and on phones are basically all regular script. 4

Drawing Things into Characters

Many of the earliest characters were "pictographs", characters drawn after the shape of things. For example, 日 ("sun") was at first a circle with a dot in the middle; 月 ("moon") was a crescent moon; 山 ("mountain") looked like three peaks joined together; and 人 ("person") looked like someone standing sideways. 5

5 慢慢地，图画变成了线条[1]，圆圈变成了方块，所以今天的“日”已经不太像太阳了。不过，如果你知道一个字原来的样子，记起来就容易多了。

6 可是，不是所有的东西都能画出来。“上”和“下”怎么画？“休息”又怎么画？于是，古人[2]想出了别的办法。一种办法是在图画上加一个记号[3]，叫“指事字[4]”。比如，在“木”的下面加一横[5]，就是“本”，本来的意思是树根；在“刀”上加一点，就是“刃[6]”，指刀最锋利[7]的地方。另[8]一种办法是把两个字放在一起，表示一个新的意思，叫“会意字[9]”。一个“人”靠在“木”旁边，就是“休”，人在树下休息；“日”和“月”放在一起，就是“明”，又亮又明白。

1 线条 *xiàntiáo* lines
2 古人 *gǔrén* people of ancient times
3 记号 *jìhao* mark, sign
4 指事字 *zhǐshìzì* indicative character
5 横 *héng* horizontal stroke
6 刃 *rèn* blade, edge
7 锋利 *fēnglì* sharp
8 另 *lìng* other, another
9 会意字 *huìyìzì* compound ideogram

Slowly the pictures became lines and the circles became squares, so today's 日 no longer looks much like the sun. But if you know what a character originally looked like, it becomes much easier to remember. 5

But not everything can be drawn. How do you draw "up" and "down"? And how do you draw "rest"? So the ancients came up with other methods. One was to add a mark to a picture; these are called "indicative characters". For example, adding a horizontal stroke at the bottom of 木 ("tree") gives 本, whose original meaning was "root"; adding a dot to 刀 ("knife") gives 刃, the sharpest part of a knife, its edge. Another was to put two characters together to express a new meaning; these are called "compound ideograms". A 人 leaning against a 木 gives 休, a person resting under a tree; 日 and 月 together give 明, bright and clear. 6

形声字[1]：一半意思，一半声音

7 象形字[2]、指事字[3]和会意字[4]加起来，其实只占汉字的一小部分。今天的汉字里，最多的是“形声字”，一般认为占百分之八十以上。形声字由两个部分组成：一部分表示意思，叫“形旁[5]”；另[6]一部分表示读音，叫“声旁[7]”。

8 举个例子：“妈”字左边是“女”，告诉我们这个字和女性[8]有关[9]；右边是“马”，告诉我们它的读音和“马”差不多。所以，“妈”不是“女的马”，而是“读音像‘马’的女性”。同样，“河”的左边是三点水[10]，表示和水有关；右边的“可[11]”原来表示读音。

1 形声字 *xíngshēngzì* phono-semantic compound
2 象形字 *xiàngxíngzì* pictograph
3 指事字 *zhǐshìzì* indicative character
4 会意字 *huìyìzì* compound ideogram
5 形旁 *xíngpáng* semantic component
6 另 *lìng* other, another
7 声旁 *shēngpáng* phonetic component
8 女性 *nǚxìng* woman, female
9 有关 *yǒuguān* to be related to
10 三点水 *sāndiǎnshuǐ* “three-dot water” radical 氵
11 可 *kě* (phonetic component here)

Phono-semantic Compounds: Half Meaning, Half Sound

Pictographs, indicative characters and compound ideograms together actually make up only a small part of all characters. Among today's characters the most numerous are phono-semantic compounds, generally thought to be more than eighty percent. A phono-semantic compound has two parts: one shows the meaning and is called the semantic component; the other shows the pronunciation and is called the phonetic component. 7

Take an example: the left side of 妈 ("mum") is 女 ("woman"), telling us that the character has to do with women; the right side is 马 ("horse"), telling us that it sounds much like 马. So 妈 is not a "female horse", but "a woman whose word sounds like 马". Likewise, the left side of 河 ("river") is the three-dot water radical, showing that it has to do with water; the 可 on the right originally showed the pronunciation. 8

9 再看一组字：清[1]、晴、请、情[2]、睛。它们的右边都是“青[3]”，读音也都跟“青”差不多。左边就不一样了：水很干净是“清”，所以有三点水[4]；出太阳的天气是“晴”，所以有“日”；请人要说话，所以有言字旁[5]；感情来自心，所以“情”有竖心旁[6]；眼睛的“睛”当然有一个“目”。

10 不过，声旁[7]并不总是可靠。汉字用了几千年，语音变化了很多，所以有的声旁和今天的读音已经差得很远了，比如“河”和“可[8]”。尽管[9]这样，学会[10]看形旁[11]和声旁，还是能帮我们猜[12]出很多生字的意思和读音。

1 清 *qīng* clear
2 情 *qíng* feeling
3 青 *qīng* blue-green
4 三点水 *sāndiǎnshuǐ* “three-dot water” radical 氵
5 言字旁 *yánzìpáng* “speech” radical 讠
6 竖心旁 *shùxīnpáng* “upright heart” radical 忄
7 声旁 *shēngpáng* phonetic component
8 可 *kě* (phonetic component here)
9 尽管 *jǐnguǎn* despite
10 学会 *xuéhuì* to learn, to master
11 形旁 *xíngpáng* semantic component
12 猜 *cāi* to guess

Now look at a group of characters: 清, 晴, 请, 情, 睛. Their right side is always 青, and they all sound much like 青. The left sides differ: clean water is 清, so it has the water radical; weather with sunshine is 晴, so it has 日; inviting someone means speaking, so 请 has the speech radical; feelings come from the heart, so 情 has the heart radical; and the 睛 of 眼睛 (“eye”) naturally has a 目 (“eye”). 9

The phonetic component, however, is not always reliable. Characters have been used for thousands of years and pronunciation has changed a great deal, so some phonetic components are now far from today’s readings, like 河 and 可. Even so, learning to read the semantic and phonetic parts helps us guess the meaning and sound of many new characters. 10

部首[1]和笔顺[2]

11 形旁[3]常常也是汉字的“部首”。部首有点儿像汉字的“姓”：带三点水[4]的字很多和水有关[5]，比如江、湖、海、洗；带“木”的字常常和树或者木头有关，比如树、林、桌；带“口”的字很多和嘴有关，比如吃、喝、叫、唱。人们查纸字典的时候，也常常先找部首，再数剩下[6]的笔画[7]。清朝[8]编的《康熙字典[9]》一共用了 214 个部首，今天，电脑里的汉字编码[10]也还是按照这些部首来排列的。

12 写汉字还要注意笔顺，也就是先写哪一笔，后写哪一笔。

1 部首 *bùshǒu* radical (of a character)
2 笔顺 *bǐshùn* stroke order
3 形旁 *xíngpáng* semantic component
4 三点水 *sāndiǎnshuǐ* “three-dot water” radical 氵
5 有关 *yǒuguān* to be related to
6 剩下 *shèngxià* to remain
7 笔画 *bǐhuà* stroke (of a character)
8 清朝 *Qīngcháo* the Qing dynasty
9 康熙字典 *Kāngxī Zìdiǎn* the Kangxi Dictionary (1716)
10 编码 *biānmǎ* character encoding

Radicals and Stroke Order

The semantic component is often also the character's "radical". A radical is 11
a bit like a character's "surname": characters with the water radical often relate to water, such as 江, 湖, 海 and 洗 ("river, lake, sea, wash"); characters with 木 often relate to trees or wood, such as 树, 林 and 桌 ("tree, forest, table"); and characters with 口 ("mouth") often relate to the mouth, such as 吃, 喝, 叫 and 唱 ("eat, drink, call, sing"). When looking something up in a paper dictionary, people also often find the radical first and then count the remaining strokes. The Kangxi Dictionary, compiled in the Qing dynasty, used 214 radicals in all, and the characters in today's computer encoding are still arranged by these radicals.

When writing characters you also need to pay attention to stroke order, that 12
is, which stroke comes first and which comes later.

12 基本的规则并不难：一般先写横[1]，再写竖[2]；先写上面，再写下面；先写左边，再写右边；有外框[3]的字，先写外面，再写里面，最后把“门”关上。比如“十”先写一横，再写一竖；“国”先写外面的框，再写里面的“玉”，最后写下面的一横。按正确的笔顺[4]写，字会写得更快、更好看；用毛笔[5]写书法[6]的时候，笔顺就更重要了。

简体字[7]和繁体字[8]

13 如果你去过中国大陆、台湾[9]和香港[10]，可能会发现那里的汉字不完全一样。中国大陆用的是简体字，新加坡[11]和马来西亚[12]的华人也多用简体字；台湾、香港和澳门[13]主要用繁体字。

1 横 *héng* horizontal stroke
2 竖 *shù* vertical stroke
3 框 *kuàng* frame
4 笔顺 *bǐshùn* stroke order
5 毛笔 *máobǐ* writing brush
6 书法 *shūfǎ* calligraphy
7 简体字 *jiǎntǐzì* simplified characters
8 繁体字 *fántǐzì* traditional characters
9 台湾 *Táiwān* Taiwan
10 香港 *Xiānggǎng* Hong Kong
11 新加坡 *Xīnjiāpō* Singapore
12 马来西亚 *Mǎláixīyà* Malaysia
13 澳门 *Àomén* Macau

The basic rules are not hard: generally horizontal before vertical, top before bottom, left before right; and for characters with an outer frame, the outside first, then the inside, and finally you "close the door". For example, 十 is written with the horizontal stroke first and then the vertical one; for 国 you write the outer frame first, then the 玉 ("jade") inside, and finally the bottom stroke. Writing in the correct order makes your characters quicker and better-looking, and when you write calligraphy with a brush, stroke order matters even more. 12

Simplified and Traditional Characters

If you have been to mainland China, Taiwan and Hong Kong, you may have noticed that the characters there are not quite the same. Mainland China uses simplified characters, as do most Chinese in Singapore and Malaysia; Taiwan, Hong Kong and Macau mainly use traditional characters. 13

14 二十世纪五十年代的时候，中国有很多人不识字[1]。为了让更多的人能读书写字，政府开始推广简化字[2]。1956年，政府公布了《汉字简化方案[3]》；1964年，又出版[4]了《简化字总表[5]》，里面一共有两千多个简化字。

15 简化[6]的方法有好几种。有的字是减少笔画[7]，比如“門[8]”变成了“门”，“馬[9]”变成了“马”；有的用了古代就有的简单写法，比如“雲[10]”变成了“云”；有的把草书[11]的写法变成了正式的字，比如“書[12]”变成了“书”；还有的是用一个字代替两个读音相同的字，比如“後[13]”和“后”现在都写成“后”。

1 识字 *shízì* to be literate
2 简化字 *jiǎnhuàzì* simplified characters
3 汉字简化方案 *Hànzì Jiǎnhuà Fāng'àn* Chinese Character Simplification Scheme
4 出版 *chūbǎn* to publish
5 简化字总表 *Jiǎnhuàzì Zǒngbiǎo* General List of Simplified Characters
6 简化 *jiǎnhuà* to simplify
7 笔画 *bǐhuà* stroke (of a character)
8 門 *mén* 门 (traditional form)
9 馬 *mǎ* 马 (traditional form)
10 雲 *yún* 云 (traditional form)
11 草书 *cǎoshū* cursive script
12 書 *shū* 书 (traditional form)
13 後 *hòu* 后 (traditional form, “after”)

In the 1950s, many people in China could not read. To help more people learn to read and write, the government began to promote simplified characters. In 1956 the government published the Chinese Character Simplification Scheme, and in 1964 it issued the General List of Simplified Characters, which contains more than two thousand of them. 14

There were several ways of simplifying. Some characters simply lost strokes: 門 became 门 (“door”), and 馬 became 马 (“horse”); some took simpler forms that had existed since ancient times, as when 雲 became 云 (“cloud”); some turned cursive handwriting into standard forms, as when 書 became 书 (“book”); and some used one character in place of two that sounded the same: 後 (“after”) and 后 (“queen”) are now both written 后. 15

16 简体字[†]好写、好学，繁体字[‡]保留了更多古代的信息，两种写法各有各的优点。其实，很多常用字在两种写法里是一样的，比如“人”“山”“大”“中”，所以学会[§]了一种，再学另[¶]一种并不太难。

要认识多少汉字？

17 汉字一共有多少个？《康熙字典[*]》收了四万七千多个字，现在最大的字典收的字已经超过了八万个。不过，这些字里大部分都很少用，很多只在古书里出现过一两次。

18 研究发现，最常用的一千个汉字，大约能占一般文章里用字的百分之九十。如果认识两千五百个常用字，就能认出一般文章里差不多百分之九十八的字。

† 简体字 *jiǎntǐzì* simplified characters
‡ 繁体字 *fántǐzì* traditional characters
§ 学会 *xuéhuì* to learn, to master
¶ 另 *lìng* other, another
* 康熙字典 *Kāngxī Zìdiǎn* the Kangxi Dictionary (1716)

Simplified characters are easier to write and learn, while traditional characters preserve more information from ancient times; each system has its own advantages. In fact many common characters are the same in both systems, such as 人, 山, 大 and 中, so once you have learned one, learning the other is not too hard. 16

How Many Characters Do You Need?

How many Chinese characters are there altogether? The Kangxi Dictionary contains over forty-seven thousand characters, and the largest dictionaries today include more than eighty thousand. But most of these are rarely used; many appear only once or twice in old books. 17

Research has found that the thousand most common characters make up about ninety percent of the characters in ordinary texts. If you know the 2,500 common characters, you can recognize almost 98 percent of the characters in an ordinary text. 18

18 中国的小学生毕业的时候，一般要认识三千个左右的汉字。

19 外国学生也有自己的目标。按照新的HSK标准，学到六级，大约要认识一千八百个汉字；学到最高的七到九级，要认识三千个。好消息是，认识了汉字以后，学词就容易多了。比如，学会[†]了“电”，再学“电话”“电脑”“电视”“电影”就快多了。汉语的词有几十万个，可是常用的汉字只有几千个，它们就像积木一样，可以组成各种各样的词。

用拼音[‡]打字

20 汉字这么多，电脑键盘[§]上却只有二十六个字母，人们是怎么打汉字的呢？今天，大部分中国人用的是拼音输入法[¶]。

† 学会 *xuéhuì* to learn, to master
‡ 拼音 *pīnyīn* pinyin romanization
§ 键盘 *jiànpán* keyboard
¶ 输入法 *shūrùfǎ* input method

Chinese primary-school pupils are generally expected to know about three thousand characters by the time they finish. 18

Foreign students have their own targets too. Under the new HSK standard, reaching level 6 takes about 1,800 characters, and reaching the top levels, 7 to 9, takes 3,000. The good news is that once you know the characters, learning words becomes much easier. For example, once you have learned 电 ("electricity"), learning 电话, 电脑, 电视 and 电影 ("telephone, computer, television, film") goes much faster. Chinese has hundreds of thousands of words but only a few thousand common characters; like building blocks, they combine into all kinds of words. 19

Typing with Pinyin

With so many characters and only twenty-six letters on a computer keyboard, how do people type Chinese? Today most Chinese people use a pinyin input method. 20

20 比如想打“你好”，就输入“nihao”，电脑会给出几个读音相同或者相近的词，你选一个就可以了。

21 汉语拼音[1]是1958年正式公布的，本来是帮助人们学习读音的工具。没想到，几十年以后，它成了人们每天打字最常用的方法。现在的输入法[2]非常聪明[3]，常常只要输入每个字的第一个字母，比如“nh”，它就能猜[4]出你想打的是“你好”。

22 不过，用拼音打字也带来了一个新问题：很多人认识字，却忘了怎么写。拿起笔来，一个很熟悉[5]的字怎么也想不起来，这种现象叫做“提笔忘字[6]”。所以，现在有不少人又开始用手写日记、练习书法[7]，让自己不忘了怎么写字。

1 拼音 *pīnyīn* pinyin romanization
2 输入法 *shūrùfǎ* input method
3 聪明 *cōngming* clever
4 猜 *cāi* to guess
5 熟悉 *shúxī* familiar
6 提笔忘字 *tíbǐ wàngzì* to pick up a pen and forget the character
7 书法 *shūfǎ* calligraphy

For example, to type 你好 ("hello") you enter "nihao"; the computer offers several words with the same or a similar sound, and you just pick one. 20

Hanyu Pinyin was officially adopted in 1958, originally as a tool to help people learn pronunciation. Unexpectedly, a few decades later it became the method people use most often to type every day. Today's input methods are very clever: often you only need to type the first letter of each character, say "nh", and it will guess that you want 你好. 21

But typing in pinyin has also brought a new problem: many people can recognize characters but have forgotten how to write them. They pick up a pen and simply cannot recall a very familiar character; this is known as "picking up the pen and forgetting the character". So quite a few people have started keeping handwritten diaries or practising calligraphy again, so as not to forget how to write. 22

结语

23 从刻在骨头上的甲骨文†，到手机屏幕‡上的拼音§输入法¶，汉字已经走过了三千多年。它们的样子变了很多次，但很多字背后的图画和道理还在。学汉字的时候，如果你多问一句“这个字为什么这样写”，你会发现，每个字都像一个小故事。

† 甲骨文 *jiǎgǔwén* oracle-bone script
‡ 屏幕 *píngmù* screen
§ 拼音 *pīnyīn* pinyin romanization
¶ 输入法 *shūrùfǎ* input method

In Closing

From oracle-bone script carved in bone to pinyin input on a phone screen, 23
Chinese characters have travelled more than three thousand years. Their shape has changed many times, but the pictures and the logic behind many characters are still there. When you learn characters, if you ask one more question, "why is this character written this way?", you will find that every character is like a little story.